MOHAMMAD ADHIL

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OBJECTIVE

Experienced Software Developer seeking an immediate job opportunity to make an impact. Open to relocate anywhere in the US on a short notice.

TECHNICAL SKILLS

Languages	C#, JavaScript, TypeScript, Python, SQL (T-SQL), HTML5, CSS3
.NET Stack	ASP.NET Core 10, Web API, Entity Framework Core, LINQ, SignalR, MediatR, xUnit, Blazor, MVC
Frontend	React, Redux Toolkit, Next.js, Tailwind CSS, Bootstrap, Axios, React Query
Databases	SQL Server, PostgreSQL, MongoDB — stored procedures, query tuning, schema design
Cloud	AWS EC2, AWS RDS (SQL Server / PostgreSQL), S3, IAM basics
AI / LLM	OpenAI API, Anthropic Claude API, RAG pipelines, Pinecone, FAISS, embeddings, prompt engineering, output guardrails, function calling
Tools	Visual Studio 2022, VS Code, Git / GitHub, Postman, Swagger / OpenAPI, JIRA, Figma, Agile / Scrum
AI Dev Tools	GitHub Copilot, Cursor, Claude Code — used daily for scaffolding, debugging, test generation

CERTIFICATIONS & LEARNING

Completed: AWS Academy Cloud Foundations · Python Programming (V Cube) · Excellence in IoT Workshop (IIT Hyderabad), Anthropic Claude 101, AI Fluency.

In Progress: Microsoft Azure Fundamentals (AZ-900) · Microsoft Azure Developer Associate (AZ-204)

PROFESSIONAL EXPERIENCE

Kotak Mahindra Bank · Mar 2024 – Aug 2024
Operations Analyst — Data, SQL & Reporting Hyderabad, India

- Developed complex SQL queries, stored procedures, Common Table Expressions (CTEs), window functions, joins, views, and indexing strategies against a core banking database to analyze and validate **10,000+ daily financial transactions**, ensuring data integrity, consistency, and regulatory compliance.
- Engineered automated data validation and reconciliation pipelines using **Python (Pandas)**, Excel VBA, and SQL, reducing manual reporting effort by **40%**, improving processing accuracy, and eliminating repetitive operational tasks.
- Designed and optimized SQL-based transaction monitoring solutions that processed **10,000+ daily banking records**, performing exception handling, anomaly detection, reconciliation, and compliance verification before end-of-day settlement.
- Built reusable SQL stored procedures, database views, parameterized queries, and reporting logic that significantly improved query performance, maintainability, scalability, and execution efficiency across operational reporting systems.
- Developed interactive operational dashboards using SQL, Excel, and reporting tools to visualize transaction volumes, reconciliation status, error trends, KPIs, and compliance metrics for executive leadership and business stakeholders.
- Created a Python automation framework that transformed a **3-hour manual reconciliation process into a fully automated 20-minute workflow**, improving productivity while minimizing human error and increasing operational efficiency.

- Investigated complex transaction discrepancies through root cause analysis, SQL debugging, historical data comparisons, and data profiling techniques, collaborating with cross-functional teams to resolve production issues and improve overall data quality.
- Identified and resolved recurring duplicate-entry defects by analyzing relational database records, tracing upstream process failures, documenting findings, and implementing corrective solutions that permanently eliminated reporting inconsistencies.
- Built and maintained a library of **15+ reusable SQL reporting templates, stored procedures, and parameterized scripts**, reducing report development time from hours to minutes while standardizing reporting across multiple business teams.
- Partnered with Compliance, Audit, Operations, and Technology teams to prepare historical datasets, exception reports, validation summaries, and audit evidence, contributing to successful quarterly audits with **zero compliance findings**.
- Converted complex regulatory and business rules into reusable SQL logic, validation frameworks, and rule-based reporting models, improving maintainability while reducing future development effort for compliance reporting.
- Designed SQL-based anomaly detection queries that proactively identified **200+ high-risk transaction exceptions**, enabling same-day investigation and significantly reducing operational risk exposure.
- Performed relational database optimization, query tuning, execution plan analysis, indexing, and performance improvements to accelerate large-volume data retrieval and reporting workloads.
- Applied software engineering principles including modular design, reusable code, version-controlled SQL scripts, documentation, debugging, testing, and continuous process improvement to develop scalable enterprise data solutions.
- Collaborated within Agile/Scrum teams, participating in requirements gathering, sprint planning, defect resolution, UAT support, and production deployments while working closely with developers, QA engineers, business analysts, and stakeholders.
- Leveraged strong programming concepts including data structures, algorithms, object-oriented programming principles, ETL processes, API-ready data modeling, and backend data processing techniques that directly align with modern **.NET, Java, Python, REST API, Microservices, Azure, AWS, and enterprise software development environments.**

IIT Hyderabad Jul 2021 – Dec 2023
Research Intern — IoT & 5G/6G Wireless Systems Hyderabad, India

- Architected and developed the research lab's primary wireless network simulation platform using **Python** and **C++**, creating a reusable framework that became the standard environment for network modeling throughout a 2.5-year research program.
- Designed modular simulation components to model **large-scale wireless communication**, network topology, packet transmission, latency, bandwidth utilization, throughput, congestion, and protocol behavior across hundreds of configurable scenarios.
- Executed and analyzed **300+ simulation experiments** involving multi-GB datasets, varying traffic loads, node densities, and network configurations, producing structured performance metrics that supported faculty research and academic publications.
- Engineered high-performance data processing pipelines in **C++** and **Python** to ingest, validate, transform, and analyze large simulation outputs, improving scalability and reducing manual data preparation.
- Identified and resolved a critical performance bottleneck that caused long-running simulations to stall on large datasets; optimized processing logic and memory usage, reducing execution time by approximately **35%**.
- Built analytical tools to process, aggregate, and visualize simulation data using **Python**, generating performance dashboards, graphs, and statistical reports used during research reviews and technical presentations.
- Independently researched and implemented the **MQTT publish-subscribe messaging protocol** to develop an IoT sensor data aggregation prototype, achieving **sub-50 ms message latency** under controlled laboratory testing.
- Applied concepts of multithreading, concurrency, asynchronous processing, event-driven programming, distributed messaging, and network communication while developing scalable simulation and data-processing workflows.

- Developed reusable automation scripts for experiment execution, result collection, log parsing, and report generation, significantly reducing repetitive manual work for researchers.
- Collaborated with faculty members, graduate researchers, and cross-functional teams to translate research objectives into software solutions, validate simulation accuracy, and communicate technical findings.
- Authored comprehensive technical documentation covering system architecture, simulation methodology, configuration management, testing procedures, assumptions, and reproducible experiment workflows.
- Presented research findings through technical reviews, explaining simulation methodology, performance analysis, and optimization results to faculty and PhD researchers.

PROFESSIONAL PROJECTS

NL2Data — Natural Language to SQL Query Engine 2026

.NET 10 · ASP.NET Core · React · OpenAI GPT-4o · SQL Server · Entity Framework Core

- Built a full-stack web app that converts plain-English business questions into executable SQL using a schema-aware GPT-4o prompt pipeline with LangChain.
- Implemented dynamic schema introspection — the backend injects live table/column metadata into the prompt context, eliminating hallucinated column names and improving query accuracy significantly.
- Added an auto-correction loop: failed queries are re-submitted with the PostgreSQL error message appended, achieving a self-healing success rate of 92% on the internal test set.
- Built a React dashboard with query history, result table preview, bar/line chart visualization, and one-click CSV export — making the tool usable for non-technical stakeholders.

FleetMind AI — Intelligent Fleet Document & Analytics Platform 2026

.NET 10 · ASP.NET Core · PostgreSQL 16 · pgvector · OpenAI GPT-4o · Tesseract.NET · EF Core 10 · Docker

- Architected an end-to-end AI document platform that ingests, OCRs, and semantically indexes 50+ weekly trucking documents (titles, fuel receipts, DOT inspections, maintenance invoices), automatically linking each to the correct truck via VIN and truck number extraction using Tesseract.NET and GPT-4o with strict JSON schema enforcement.
- Designed a hybrid AI query engine that classifies natural language questions as SQL, RAG, or hybrid using GPT-4o-mini; SQL route generates schema-aware parameterized queries via GPT-4o (Dapper, read-only PostgreSQL role), RAG route retrieves document chunks via pgvector HNSW cosine similarity (1536-dim embeddings) and synthesizes grounded answers with mandatory citations.
- Engineered 7-layer hallucination prevention: strict JSON schema extraction (additionalProperties: false), RAG grounding prompts, 0.72 cosine similarity threshold gate, read-only PostgreSQL role, SQL keyword security guard, execution-grounded answer formatting, and mandatory citation arrays on every response.
- Built an asynchronous document processing pipeline (IHostedService + Channel<T>) covering OCR, GPT-4o entity extraction, 512-token chunking with 64-token overlap, batch embedding via text-embedding-3-small (1536d), and pgvector storage — upload to queryable in under 30 seconds.

EDUCATION

M.S. in Information Systems Central Michigan University Aug 2024 – May 2026
Mt. Pleasant, MI

B. Tech in Electronics & Communication Engineering Jul 2020 – Jun 2024
Hyderabad, India